

Program : Diploma in Manufacturing Technology	
Course Code : 4709	Course Title: Mechanical Engineering Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide students to get a good knowledge about the importance of studying Mechanical engineering in daily life and industry.
- To familiarize with the basics of thermal engineering.
- To familiarize with the basics of Refrigeration and Air conditioning

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Science		Mathematics I & II	1&2
Basic Science		Applied Physics I & II	1 & 2
Program Core Course		Mechanical Engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize the working and performance testing of IC engines.	15	Applying
CO	Determine the fuel properties such as viscosity, calorific value, flash point, fire point of fuels.	9	Applying
CO3	Familiarize with the working and performance testing of air compressor	9	Applying

CO4	Identify the components of refrigerator and conduct experiments on refrigeration test rig	9	Understanding
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2	1			2			
CO3				2			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working and systems of Diesel and petrol engines and conduct various performance tests		
M1.01	Study the parts and functions of the petrol engine	3	Understanding
M1.02	Conduct different tests on petrol engine – load test, Morse test.	4	Applying
M1.03	Study the parts and functions of diesel engine.	4	Understanding
M1.04	Conducting different tests on diesel engine – load test.	4	Applying
Contents: Study of petrol and Diesel engines. Components and systems of IC engines - Load test on petrol and diesel engines.			
CO2	Distinguish the terms viscosity, calorific value, flash point, fire point of fuels.		
M2.01	Perform the test on lubricating oils.Determine the viscosity of the given oil.	3	Applying
M2.02	Determine the flash and fire point of the given oil.	3	Applying

M2.03	Determine the calorific value of given solid fuels. Determine the calorific value of the given liquid fuel.	3	Applying
	Lab Exam – I	1.5	
Contents: Properties of fuels, Redwood Viscometer. Flash and fire point apparatus. Bomb Calorimeters Junkers calorimeter.			
CO3	Describe the working of air compressor and conduct the performance test		
M3.01	Study the components and working of Reciprocating and Rotary air compressors	3	Understanding
M3.02	Conduct a performance test on air compressor and determine volumetric efficiency, isothermal efficiency & adiabatic efficiency	6	Applying
Contents: Air compressor –Components – working-Reciprocating and Rotary air compressors- Performance test, volumetric efficiency, adiabatic efficiency.			
CO4	Experiments on refrigeration test rig, components of refrigerator.		
M 4.01	Study of domestic refrigerator	2	Understanding
M 4.02	To conduct test to find COP and capacity of given refrigeration test rig	3	Understanding
M 4.03	To conduct COP test on given water cooler	4	Understanding
	Lab Exam – II	1.5	
Contents: Experiments on refrigeration test rig, components of refrigerator			

Text / Reference

T/R	Book Title/Author
T1	Mechanical Workshop & Laboratory Manual By K. C. John
R1	Thermal engineering by R S Khurmi
R2	Internal combustion engines by V Ganesan
R3	Refrigeration and air conditioning by C P Arora
R4	A textbook of automobile engineering by S K Gupta

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/112/105/112105129/
2	https://nptel.ac.in/courses/107/106/107106088/
3	https://lecturenotes.in/video-tutorial/15813-note-of-thermal-engineering-1-video-tutorial-by-bhaktishree-sathpathy
4	https://onlinecourses.nptel.ac.in/noc20_me71/preview